

STANDARD 2

Stage 6

Algebra (Std 2), A4 Types of Relationships (Y12)

Non-Linear: Exponential/Quadratics (Std 2)

Teacher:

Exam Equivalent Time: 141 minutes (based on allocation of 1.5 minutes per mark)

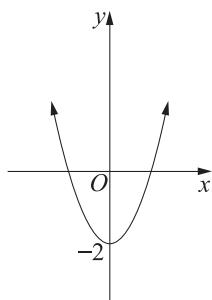


Questions

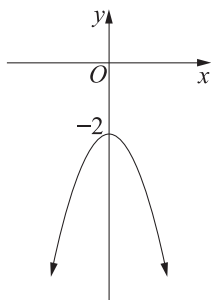
1. Algebra, STD2 A4 2018 HSC 4 MC

Which graph best represents the equation $y = x^2 - 2$?

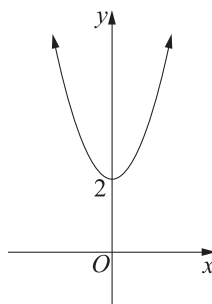
A.



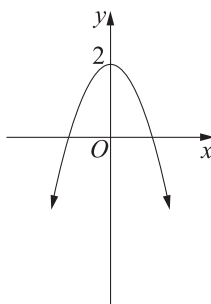
B.



C.

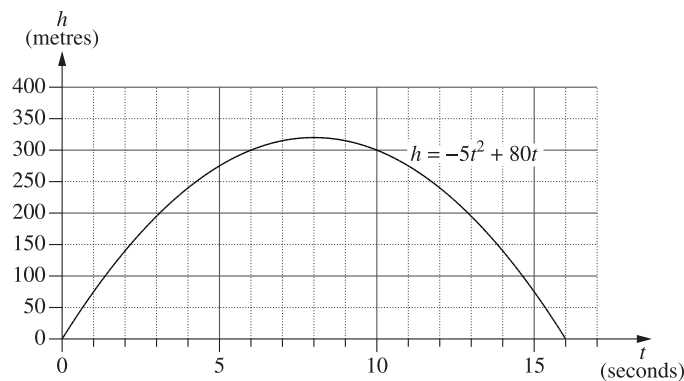


D.



2. Algebra, STD2 A4 2022 HSC 9 MC

An object is projected vertically into the air. Its height, h metres, above the ground after t seconds is given by $h = -5t^2 + 80t$.



For how long is the object at a height of 300 metres or more above the ground?

- A. 4 seconds
- B. 6 seconds
- C. 8 seconds
- D. 10 seconds

3. Algebra, STD2 A4 2006 HSC 14 MC

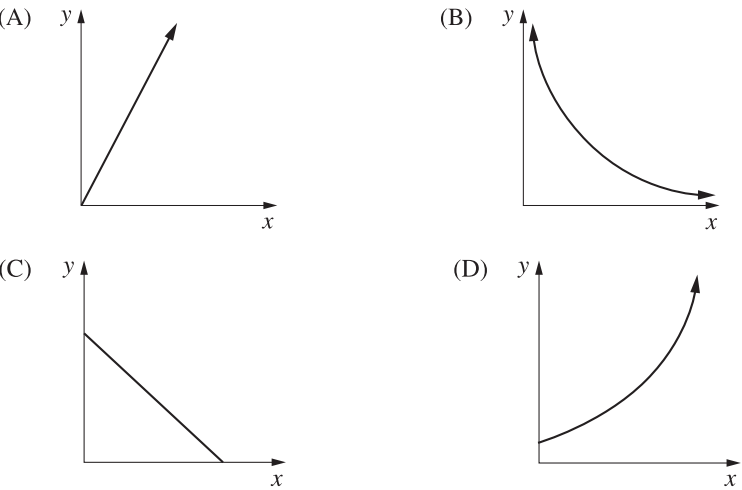
In 2004 there were 13.5 million registered motor vehicles in Australia. The number of registered motor vehicles is increasing at a rate of 2.3% per year.

Which expression represents the number (in millions) of registered motor vehicles, if y represents the number of years after 2004?

- A. $13.5 \times (1.023)^y$
- B. $13.5 \times (0.023)^y$
- C. $13.5 \times (1.023) \times y$
- D. $13.5 \times (0.023) \times y$

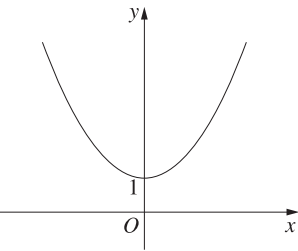
4. Algebra, STD2 A4 2008 HSC 4 MC

Which graph best represents $y = 3^x$?



5. Algebra, STD2 A4 2014 HSC 3 MC

The diagram shows the graph of an equation.

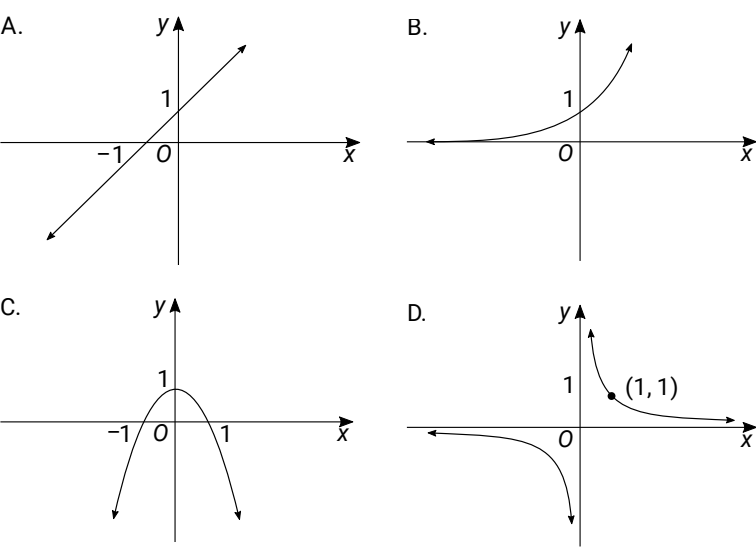


Which of the following equations does the graph best represent?

- A. $y = \frac{3}{x} + 1$
- B. $y = 3^x + 1$
- C. $y = 3x^2 + 1$
- D. $y = 3x^3 + 1$

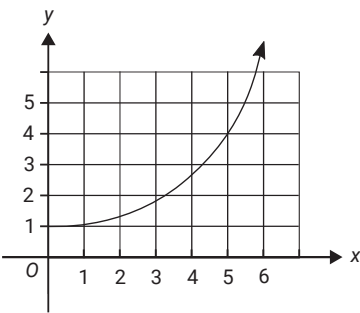
6. Algebra, STD2 A4 2020 HSC 1 MC

Which of the following could represent the graph of $y = -x^2 + 1$?



7. Algebra, STD2 A4 SM-Bank 1 MC

The graph below represents $y = a^x$.

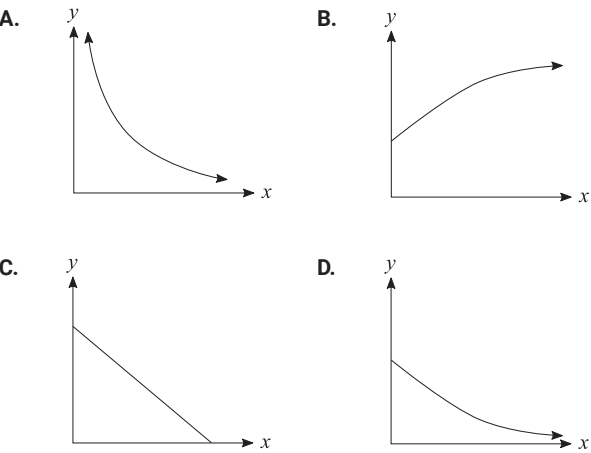


What is the approximate value of $2(a^5)$?

- A. 4
- B. 8
- C. 32
- D. 64

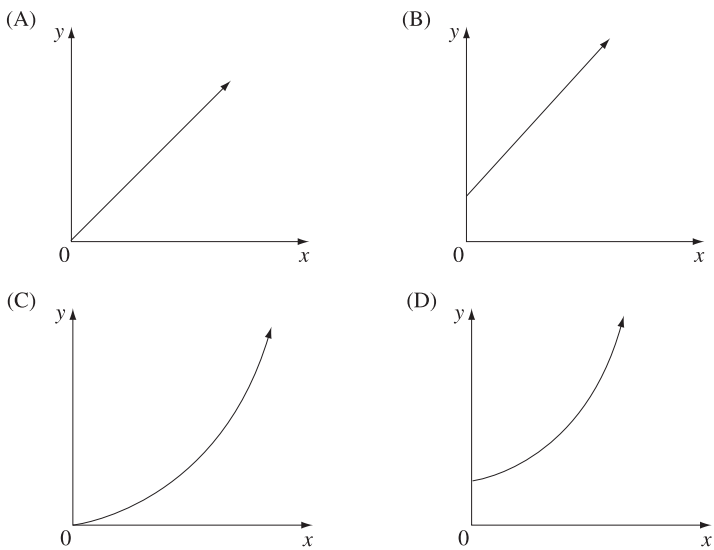
8. Algebra, STD2 A4 SM-Bank 3 MC

Which graph best represents the equation $y = 5^{-x}$?



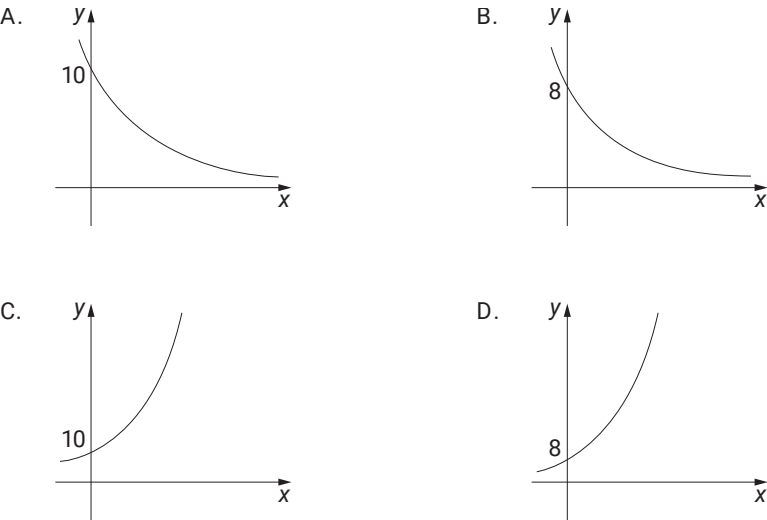
9. Algebra, STD2 A4 2011 HSC 6 MC

Which of the following graphs best represents the equation $y = a^x$, where a is a positive number greater than 1?



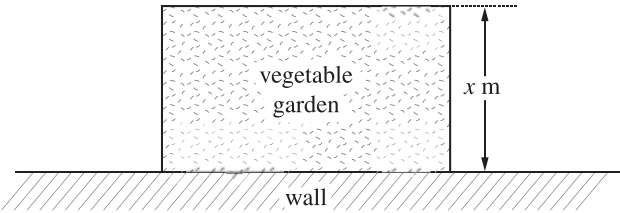
10. Algebra, STD2 A4 2021 HSC 10 MC

Which of the following best represents the graph of $y = 10(0.8)^x$?



11. Algebra, STD2 A4 2013 HSC 22 MC

Leanne wants to build a rectangular vegetable garden in her backyard. She has 20 metres of fencing and will use a wall as one side of the garden. The plan for her garden is shown, where x metres is the width of her garden.



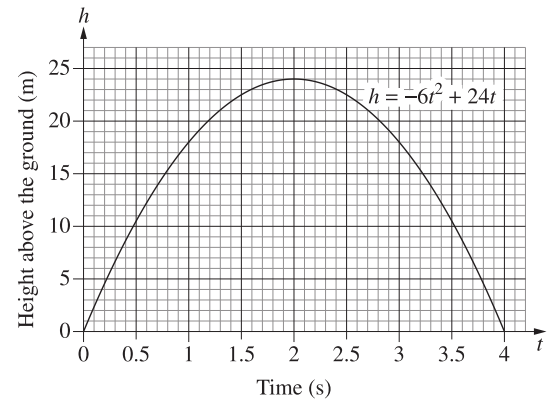
Which equation gives the area, A , of the vegetable garden?

- A. $A = 10x - x^2$
- B. $A = 10x - 2x^2$
- C. $A = 20x - x^2$
- D. $A = 20x - 2x^2$

12. Algebra, STD2 A4 2023 HSC 20

On another planet, a ball is launched vertically into the air from the ground.

The height above the ground, h metres, can be modelled using the function $h = -6t^2 + 24t$, where t is measured in seconds. The graph of the function is shown.



- a. Based on the graph, what is the maximum height reached by the ball? (1 mark)
- b. Based on the graph, at what TWO times is the ball at $\frac{3}{4}$ of its maximum height? (2 marks)

13. Algebra, STD2 A4 2011 HSC 26b

Jack needs to find the number of years, t , it will take for a population of bats to first exceed 18 000.

He uses a ‘guess-and-check’ method to estimate t in the following equation

$5 \times 3^t = 18\,000.$

Here is his working:

Try $t = 9$

$5 \times 3^9 = 98\,415$

Conclusion: $t = 9$ is too big.

- i. Jack’s next guess is $t = 6$. Show Jack’s correct working for this guess, including the calculation and conclusion. (1 mark)
- ii. Continue using the ‘guess-and-check’ method to find the number of years, t , it will take for the population to first exceed 18 000, if t is a whole number. Include the calculations and conclusions. (2 marks)

14. Algebra, STD2 A4 2023 HSC 22

The braking distance of a car, in metres, is directly proportional to the square of its speed in km/h, and can be represented by the equation

$\text{braking distance} = k \times (\text{speed})^2$

where k is the constant of variation.

The braking distance for a car travelling at 50 km/h is 20 m.

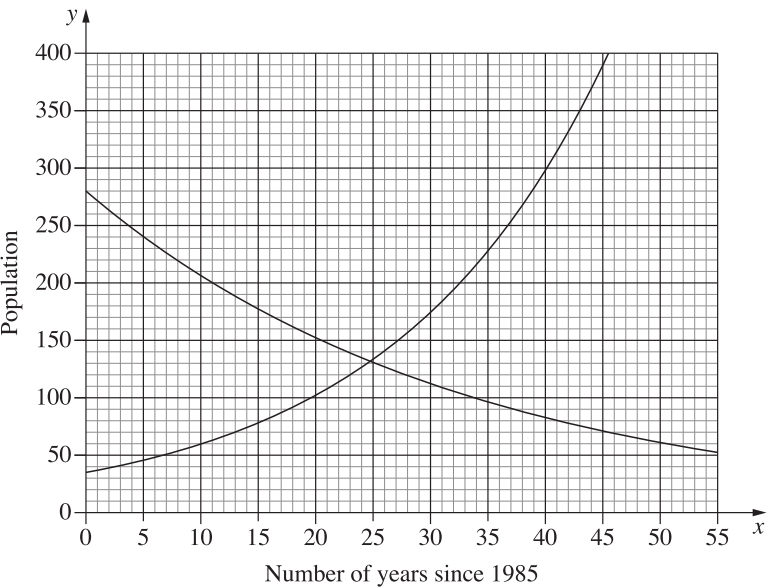
- a. Find the value of k . (2 marks)
- b. What is the braking distance when the speed of the car is 90 km/h? (1 mark)

15. Algebra, STD2 A4 2024 HSC 22

The graph shows the populations of two different animals, W and K , in a conservation park over time. The y -axis is the size of the population and the x -axis is the number of years since 1985 .

Population W is modelled by the equation $y = A \times (1.055)^x$.

Population K is modelled by the equation $y = B \times (0.97)^x$.



Complete the table using the information provided. (3 marks)

	Population W	Population K
Population in 1985	$A = 34$	$B =$
Percentage yearly change in the population		
Predicted population when $x = 50$		61

16. Algebra, STD2 A4 SM-Bank 2

Moses finds that for a Froghead eel, its mass is directly proportional to the square of its length.

An eel of this species has a length of 72 cm and a mass of 8250 grams.

What is the expected length of a Froghead eel with a mass of 10.2 kg? Give your answer to one decimal place. **(3 marks)**

17. Algebra, STD2 A4 2013 HSC 30a

Wind turbines, such as those shown, are used to generate power.



Acknowledgement: Stock Photo–Wind Turbines – Image ID: 1051412 © Miguel Saavedra

In theory, the power that could be generated by a wind turbine is modelled using the equation

$$T = 20\,000w^3$$

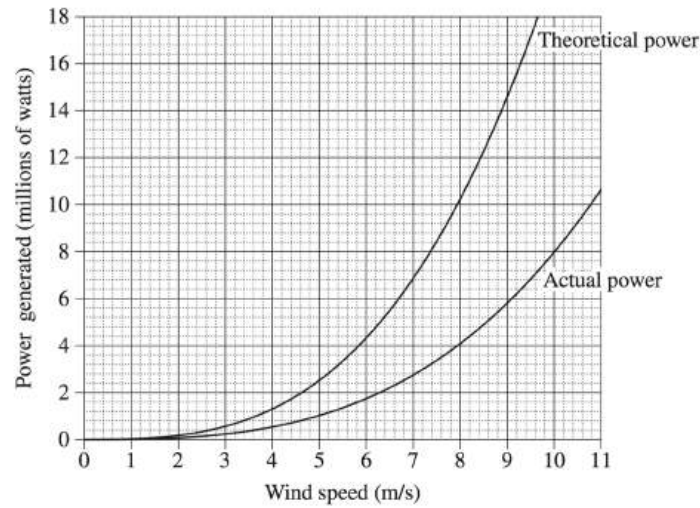
where T is the theoretical power generated, in watts
 w is the speed of the wind, in metres per second.

i. Using this equation, what is the theoretical power generated by a wind turbine if the wind speed is 7.3 m/s? (1 mark)

In practice, the actual power generated by a wind turbine is only 40% of the theoretical power.

ii. If A is the actual power generated, in watts, write an equation for A in terms of w . (1 mark)

The graph shows both the theoretical power generated and the actual power generated by a particular wind turbine.



iii. Using the graph, or otherwise, find the difference between the theoretical power and the actual power generated when the wind speed is 9 m/s. (1 mark)

A particular farm requires at least 4.4 million watts of actual power in order to be self-sufficient.

iv. What is the minimum wind speed required for the farm to be self-sufficient? (1 mark)

A more accurate formula to calculate the power (P) generated by a wind turbine is

$$P = 0.61 \times \pi \times r^2 \times w^3$$

where r is the length of each blade, in metres

w is the speed of the wind, in metres per second.

Each blade of a particular wind turbine has a length of 43 metres. The turbine operates at a wind speed of 8 m/s.

v. Using the formula above, if the wind speed increased by 10%, what would be the percentage increase in the power generated by this wind turbine? (3 marks)

18. Algebra, STD2 A4 2004 HSC 26a

i. The number of bacteria in a culture grows from 100 to 114 in one hour.

What is the percentage increase in the number of bacteria? (1 mark)

ii. The bacteria continue to grow according to the formula $n = 100(1.14)^t$, where n is the number of bacteria after t hours.

What is the number of bacteria after 15 hours? (1 mark)

Time in hours (t)	0	5	10	15
Number of bacteria (n)	100	193	371	?

iii. Use the values of n from $t = 0$ to $t = 15$ to draw a graph of $n = 100(1.14)^t$.

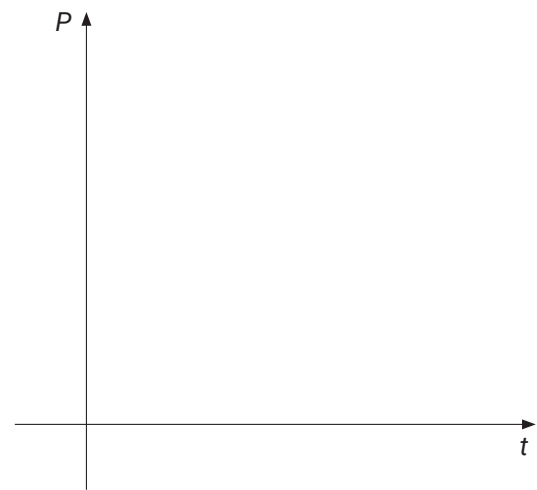
Use about half a page for your graph and mark a scale on each axis. (4 marks)

iv. Using your graph or otherwise, estimate the time in hours for the number of bacteria to reach 300. (1 mark)

19. Algebra, STD2 A4 2021 HSC 24

A population, P , is to be modelled using the function $P = 2000(1.2)^t$, where t is the time in years.

- a. What is the initial population? (1 mark)
- b. Find the population after 5 years. (1 mark)
- c. On the axes below, draw the graph of the population against time, showing the points at $t = 0$ and at $t = 5$. (2 marks)



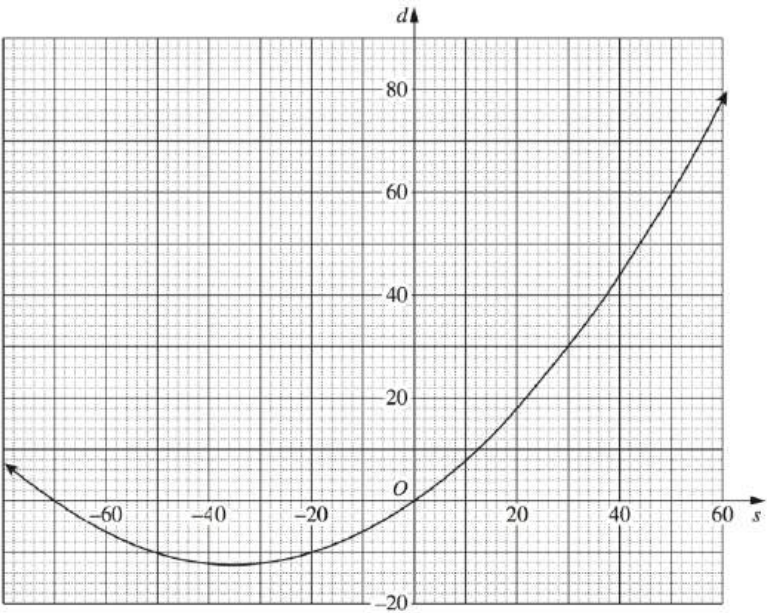
20. Algebra, STD2 A4 2009 HSC 28a

Anjali is investigating stopping distances for a car travelling at different speeds. To model this she uses the equation

$$d = 0.01s^2 + 0.7s,$$

where d is the stopping distance in metres and s is the car's speed in km/h.

The graph of this equation is drawn below.



- i. Anjali knows that only part of this curve applies to her model for stopping distances. In your writing booklet, using a set of axes, sketch the part of this curve that applies for stopping distances. (1 mark)
- ii. What is the difference between the stopping distances in a school zone when travelling at a speed of 40 km/h and when travelling at a speed of 70 km/h? (2 marks)

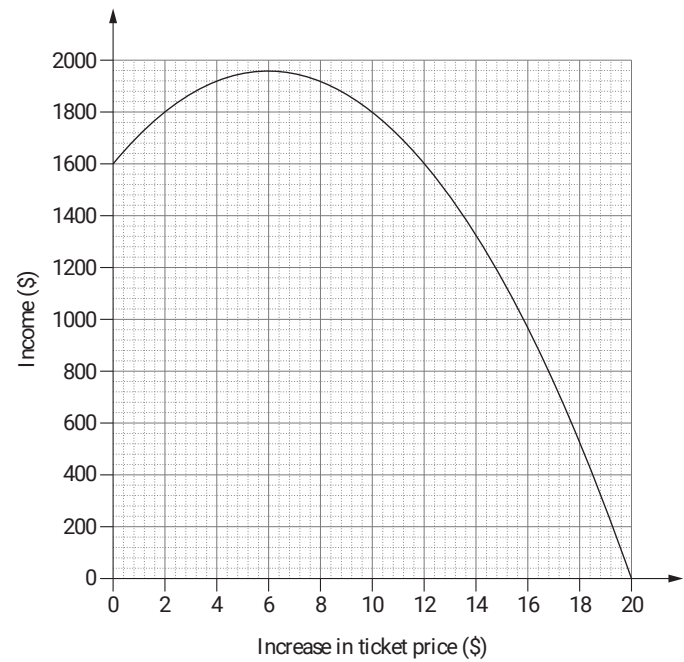
21. Algebra, STD2 A4 2017 HSC 28e

A movie theatre has 200 seats. Each ticket currently costs \$8.

The theatre owners are currently selling all 200 tickets for each session. They decide to increase the price of tickets to see if they can increase the income earned from each movie session.

It is assumed that for each one dollar increase in ticket price, there will be 10 fewer tickets sold.

A graph showing the relationship between an increase in ticket price and the income is shown below.



- i. What ticket price should be charged to maximise the income from a movie session? (1 mark)
- ii. What is the number of tickets sold when the income is maximised? (1 mark)
- iii. The cost to the theatre owners of running each session is \$500 plus \$2 per ticket sold.
Calculate the profit earned by the theatre owners when the income earned from a session is maximised. (2 marks)

22. Algebra, STD2 A4 2009 HSC 28c

The height above the ground, in metres, of a person's eyes varies directly with the square of the distance, in kilometres, that the person can see to the horizon.

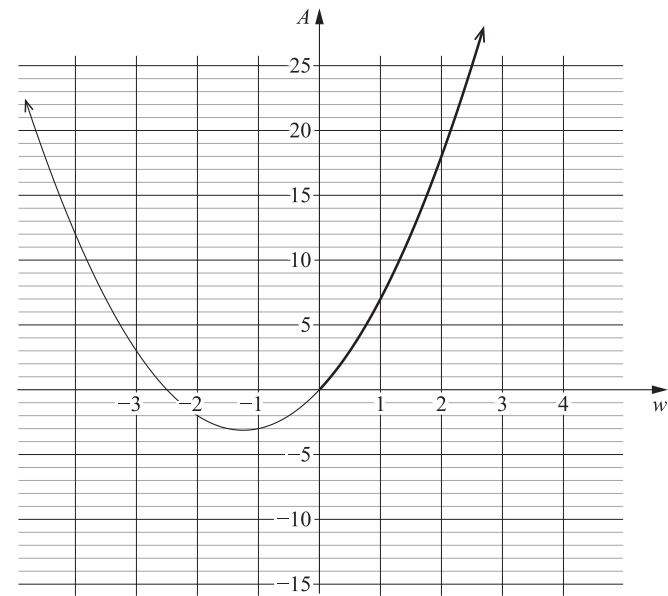
A person whose eyes are 1.6 m above the ground can see 4.5 km out to sea.

How high above the ground, in metres, would a person's eyes need to be to see an island that is 15 km out to sea? Give your answer correct to one decimal place. (3 marks)

23. Algebra, STD2 A4 2019 HSC 31

A rectangle has width w centimetres. The area of the rectangle, A , in square centimetres, is $A = 2w^2 + 5w$.

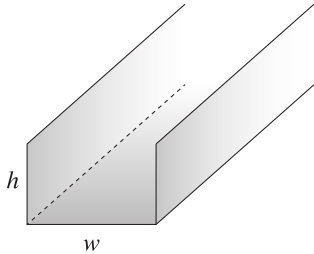
The graph $A = 2w^2 + 5w$ is shown.



- a. Explain why, in this context, the model $A = 2w^2 + 5w$ only makes sense for the bold section of the graph. (1 mark)
- b. The area of the rectangle is 18 cm². Calculate the perimeter of the rectangle. (2 marks)

24. Algebra, STD2 A4 2024 HSC 26

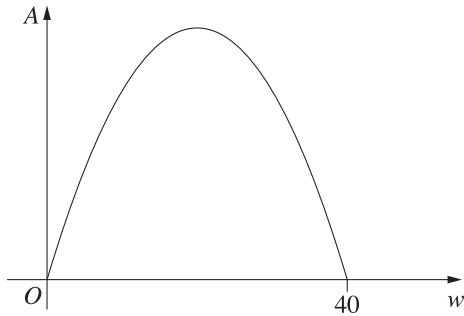
A sheet of metal is folded to make a gutter, as shown. The cross-section of the gutter is a rectangle of width w cm and height h cm.



The area, A cm², of the cross-section can be modelled by the quadratic formula

$$A = -0.5w^2 + 20w$$

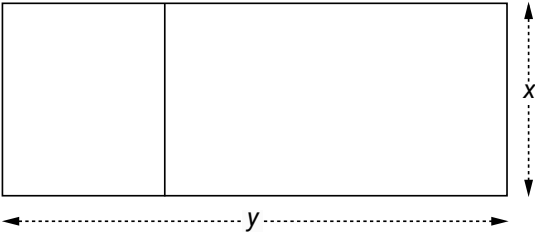
A graph of this model is shown.



Find the width and height of the rectangle which will give the greatest possible area of the cross-section. (3 marks)

25. Algebra, STD2 A4 2020 HSC 19

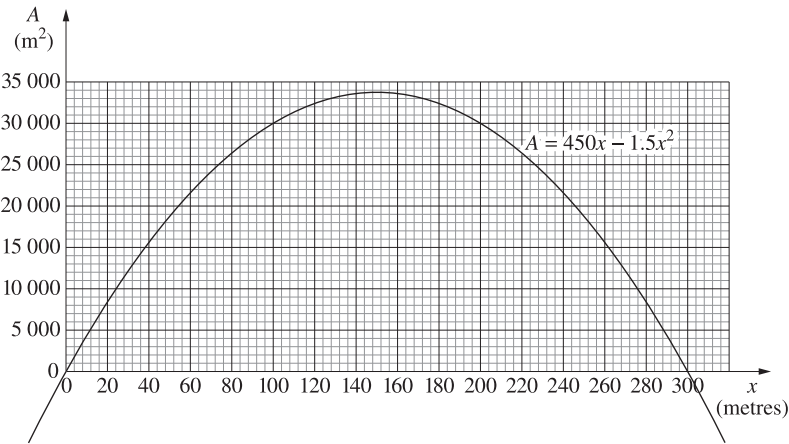
A fence is to be built around the outside of a rectangular paddock. An internal fence is also to be built. The side lengths of the paddock are x metres and y metres, as shown in the diagram.



A total of 900 metres of fencing is to be used. Therefore $3x + 2y = 900$.

The area, A , in square metres, of the rectangular paddock is given by $A = 450x - 1.5x^2$.

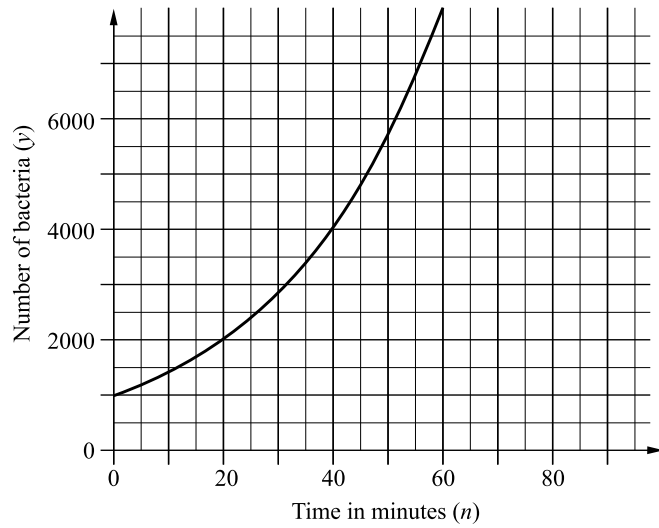
The graph of this equation is shown.



- a. If the area of the paddock is $30\,000\text{ m}^2$, what is the largest possible value of x ? (1 mark)
- b. Find the values of x and y so that the area of the paddock is as large as possible. (2 marks)
- c. Using your value from part (b), find the largest possible area of the paddock. (1 mark)

26. Algebra, STD2 A4 2020 HSC 33

The graph shows the number of bacteria, y , at time n minutes. Initially (when $n = 0$) the number of bacteria is 1000.

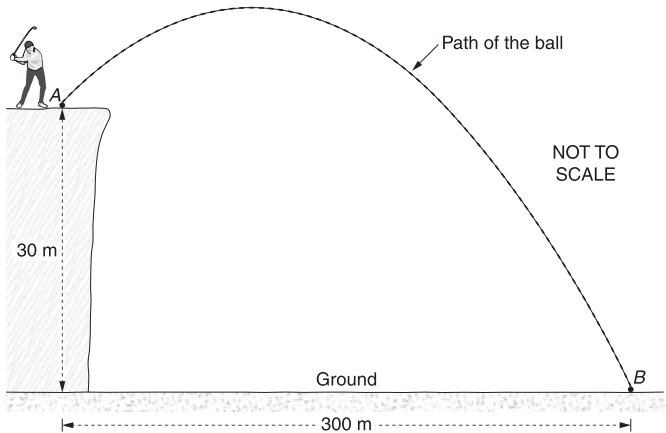


a. Find the number of bacteria at 40 minutes. **(1 mark)**

b. The number of bacteria can be modelled by the equation $y = A \times b^n$, where A and b are constants. Use the guess and check method to find, to two decimal places, an upper and lower estimate for the value of b . The upper and lower estimates must differ by 0.01. **(2 marks)**

27. Algebra, STD2 A4 2012 HSC 30b

A golf ball is hit from point **A** to point **B**, which is on the ground as shown. Point **A** is 30 metres above the ground and the horizontal distance from point **A** to point **B** is 300 m.



The path of the golf ball is modelled using the equation

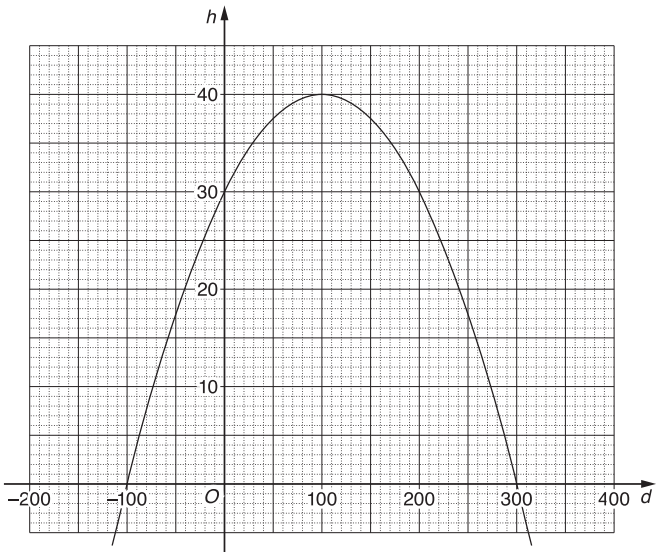
$$h = 30 + 0.2d - 0.001d^2$$

where

h is the height of the golf ball above the ground in metres, and

d is the horizontal distance of the golf ball from point **A** in metres.

The graph of this equation is drawn below.



i. What is the maximum height the ball reaches above the ground? (1 mark)

ii. There are two occasions when the golf ball is at a height of 35 metres.

What horizontal distance does the ball travel in the period between these two occasions? (1 mark)

iii. What is the height of the ball above the ground when it still has to travel a horizontal distance of 50 metres to hit the ground at point **B**? (1 mark)

iv. Only part of the graph applies to this model.

Find all values of **d** that are not suitable to use with this model, and explain why these values are not suitable. (2 marks)

28. Algebra, STD2 A4 2012 HSC 30c

In 2010, the city of Thagoras modelled the predicted population of the city using the equation

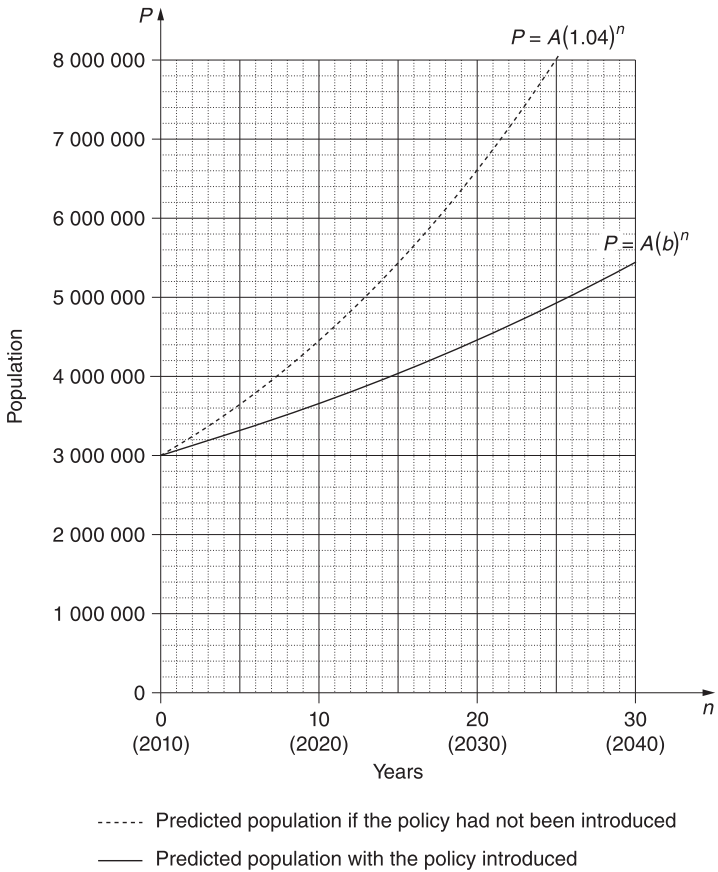
$$P = A(1.04)^n.$$

That year, the city introduced a policy to slow its population growth. The new predicted population was modelled using the equation

$$P = A(b)^n.$$

In both equations, P is the predicted population and n is the number of years after 2010.

The graph shows the two predicted populations.



- i. Use the graph to find the predicted population of Thagoras in 2030 if the population policy had NOT been introduced. (1 mark)
- ii. In each of the two equations given, the value of A is 3 000 000.
What does A represent? (1 mark)
- iii. The guess-and-check method is to be used to find the value of b , in $P = A(b)^n$.
(1) Explain, with or without calculations, why 1.05 is not a suitable first estimate for b . (1 mark)
(2) With $n = 20$ and $P = 4\,460\,000$, use the guess-and-check method and the equation $P = A(b)^n$ to estimate the value of b to two decimal places. Show at least TWO estimate values for b , including

calculations and conclusions. (2 marks)

- iv. The city of Thagoras was aiming to have a population under 7 000 000 in 2050. Does the model indicate that the city will achieve this aim?

Justify your answer with suitable calculations. (2 marks)

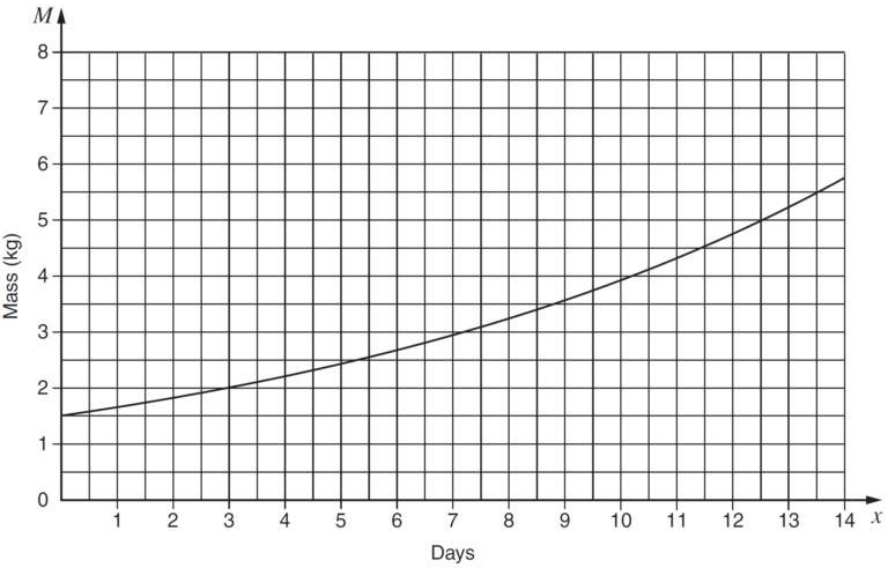
29. Algebra, STD2 A4 2006 HSC 28b

A new tunnel is built. When there is no toll to use the tunnel, 6000 vehicles use it each day. For each dollar increase in the toll, 500 fewer vehicles use the tunnel.

- i. Find the lowest toll for which no vehicles will use the tunnel. (1 mark)
- ii. For a toll of \$5.00, how many vehicles use the tunnel each day and what is the total daily income from tolls? (2 marks)
- iii. If d (dollars) represents the value of the toll, find an equation for the number of vehicles (v) using the tunnel each day in terms of d . (2 marks)
- iv. Anne says 'A higher toll always means a higher total daily income'.
Show that Anne is incorrect and find the maximum daily income from tolls. (Use a table of values, or a graph, or suitable calculations.) (3 marks)

30. Algebra, STD2 A4 2016 HSC 29b

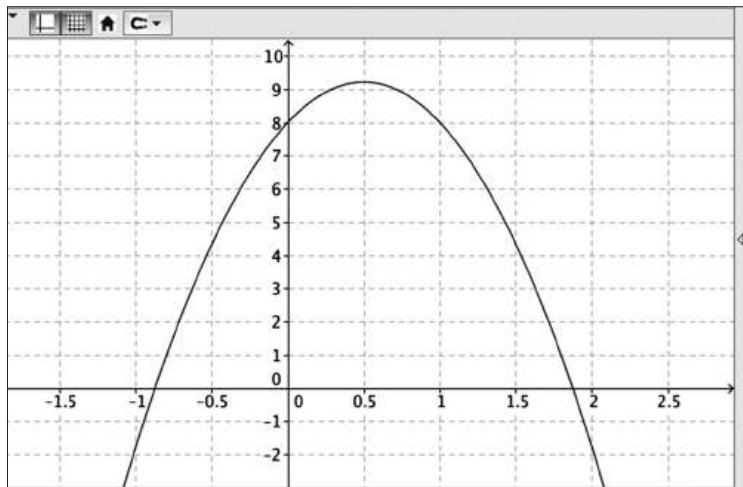
The mass M kg of a baby pig at age x days is given by $M = A(1.1)^x$ where A is a constant. The graph of this equation is shown.



- i. What is the value of A ? (1 mark)
- ii. What is the daily growth rate of the pig's mass? Write your answer as a percentage. (1 mark)

31. Algebra, STD2 A4 2015 HSC 29e

A diver springs upwards from a diving board, then plunges into the water. The diver's height above the water as it varies with time is modelled by a quadratic function. Graphing software is used to produce the graph of this function.



Explain how the graph could be used to determine how high above the height of the diving board the diver was when he reached the maximum height. **(2 marks)**

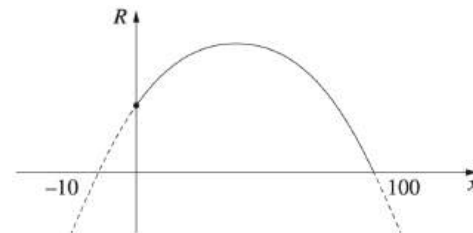
32. Algebra, STD2 A4 2021 HSC 35

A publisher sells a book for \$10. At this price, 5000 copies of the book will be sold and the revenue raised will be $5000 \times 10 = \$50\,000$.

The publisher is considering increasing the price of the book. For every dollar the price of the book is increased, the publisher will sell 50 fewer copies of the book.

If the publisher charges $(10 + x)$ dollars for each book, a quadratic model for the revenue raised, R , from selling books is

$$R = -50x^2 + 4500x + 50\,000$$



- By first finding a suitable value of x , find the price the publisher should charge for each book to maximise the revenues raised from sales of the book. **(2 marks)**
- Find the value of the intercept of the parabola with the vertical axis. **(1 mark)**

STANDARD 2

Stage 6

Algebra (Std 2), A4 Types of Relationships (Y12)

Non-Linear: Exponential/Quadratics (Std 2)

Teacher:

Exam Equivalent Time: 141 minutes (based on allocation of 1.5 minutes per mark)



Worked Solutions

- Algebra, STD2 A4 2018 HSC 4 MC

$$y = x^2 - 2$$

y -intercept = -2 (when $x = 0$)

Quadratic is positive with vertex at $y = -2$

$\Rightarrow A$

- Algebra, STD2 A4 2022 HSC 9 MC

Object reaches 300 m when $t = 6$ seconds.

Object drops back below 300 m when $t = 10$ seconds.

Time at 300 m or above = $10 - 6 = 4$ seconds

$\Rightarrow A$

- Algebra, STD2 A4 2006 HSC 14 MC

In 2004, 13.5 million

$$1 \text{ year later} = 13.5 \times (1.023)$$

$$2 \text{ years later} = 13.5 \times (1.023) \times (1.023)$$

$$= 13.5 \times (1.023)^2$$

$$\therefore y \text{ years later} = 13.5 \times (1.023)^y$$

$\Rightarrow A$

- Algebra, STD2 A4 2008 HSC 4 MC

$y = 3^x$ passes through $(0, 1)$ and is exponential.

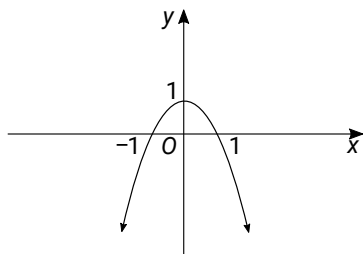
$\Rightarrow D$

- Algebra, STD2 A4 2014 HSC 3 MC

Graph is a parabola that passes through $(0, 1)$.

$\Rightarrow C$

6. Algebra, STD2 A4 2020 HSC 1 MC



♦ Mean mark 53%.
COMMENT: A surprisingly low State mean mark!

$\Rightarrow C$

7. Algebra, STD2 A4 SM-Bank 1 MC

From graph:

$$a^5 = 4$$

$$\therefore 2(a^5) = 2 \times 4$$

$$= 8$$

$\Rightarrow B$

8. Algebra, STD2 A4 SM-Bank 3 MC

$$y = 5^{-x}$$

$$= \frac{1}{5^x}$$

$$\text{At } x = 0, y = 1$$

$$\text{As } x \rightarrow \infty, y \rightarrow 0$$

$\Rightarrow D$

9. Algebra, STD2 A4 2011 HSC 6 MC

$$\text{At } x = 0, y = 1$$

$$\text{As } x \uparrow, y \uparrow \text{ exponentially}$$

$\Rightarrow D$

♦ Mean mark 48%

10. Algebra, STD2 A4 2021 HSC 10 MC

By elimination:

$$\text{When } x = 0, y = 10(0.8)^0 = 10$$

$\rightarrow$ Eliminate B and D

$$\text{As } x \rightarrow \infty, y \rightarrow 0$$

$\rightarrow$ Eliminate C

$\Rightarrow A$

♦ Mean mark 41%.

11. Algebra, STD2 A4 2013 HSC 22 MC

$$\text{Length of garden} = (20 - 2x)$$

$$\text{Area} = x(20 - 2x)$$

$$= 20x - 2x^2$$

$$\Rightarrow D$$

♦♦♦ Mean mark 24% (lowest mean of any MC question in 2013 exam)

12. Algebra, STD2 A4 2023 HSC 20

a. $h_{\max} = 24$ metres

b. $\frac{3}{4} \times h_{\max} = \frac{3}{4} \times 24 = 18$ metres

From graph, ball is at at 18 metres when:

$t = 1$ and 3 seconds

13. Algebra, STD2 A4 2011 HSC 26b

i. When $t = 6$,

$$5 \times 3^6 = 3645$$

$\Rightarrow$ Too small

ii. When $t = 7$,

$$5 \times 3^7 = 10\,935$$

$\Rightarrow$ Too small

When $t = 8$,

$$5 \times 3^8 = 32\,805$$

$\Rightarrow$ exceeds 18 000

$\therefore t = 8$

14. Algebra, STD2 A4 2023 HSC 22

a. $\text{braking distance} = k \times (\text{speed})^2$

$$20 = k \times 50^2$$

$$k = \frac{20}{50^2}$$

$$= 0.008$$

b. Find braking distance (d) when speed = 90 km/h:

$$d = 0.008 \times 90^2$$

$$= 64.8 \text{ m}$$

15. Algebra, STD2 A4 2024 HSC 22

	Population <i>W</i>	Population <i>K</i>
Population in 1985	<i>A</i> = 34	<i>B</i> = 280
Percentage yearly change in the population	+ 5.5%	− 3.0%
Predicted population when <i>x</i> = 50	494	61

Calculations:
Population *W*: $(1.055)^x \Rightarrow$ increase of 5.5% each year
Population *K*: $(0.97)^x \Rightarrow 1 - 0.97 = 0.03 \Rightarrow$ decrease of 3.0% each year
Predicted population (*W*) = $34 \times (1.055)^{50} = 494$

16. Algebra, STD2 A4 SM-Bank 2

Mass $\propto$ length²
 $m = kl^2$

Find *k*:
 $8250 = k \times 72^2$
 $k = \frac{8250}{72^2}$
 $= 1.591 \dots$

When *l* when *m* = 10 200:
 $10\,200 = 1.591 \dots \times l^2$
 $l^2 = \frac{10\,200}{1.591 \dots}$
 $\therefore l = 80.069 \dots$
 $= 80.1 \text{ cm (to 1 d.p.)}$

17. Algebra, STD2 A4 2013 HSC 30a

i. $T = 20\,000w^3$

If $w = 7.3$

$$T = 20\,000 \times (7.3)^3$$
$$= 7\,780\,340 \text{ watts}$$

ii. We know $A = 40\% \times T$

$$\Rightarrow A = 0.4 \times 20\,000 \times w^3$$
$$= 8000w^3$$

iii. Solution 1

At $w = 9$

$A = 5.8$ million watts (from graph)

$T = 14.6$ million watts (from graph)

Difference = 14.6 million – 5.8 million

=8.8 million watts

Alternative Solution

At $w = 9$

$$T = 20\,000 \times 9^3$$
$$= 14\,580\,000 \text{ watts}$$

$$A = 8000 \times 9^3$$
$$= 5\,832\,000 \text{ watts}$$

Difference = 14 580 000 – 5 832 000

= 8 748 000 watts

iv. Find w if $A = 4.4$ million

$$8000w^3 = 4\,400\,000$$

$$w^3 = \frac{4\,400\,000}{8000}$$
$$= 550$$

$$\therefore w = \sqrt[3]{550}$$
$$= 8.1932\dots$$
$$= 8.2 \text{ m/s (1 d.p.)}$$

$\therefore$ The minimum wind speed required is 8.2 m/s

Find P when $w = 8$ and $r = 43$

♦ Mean mark 34%

♦ Mean mark 38%

♦♦ Mean mark 25%
COMMENT: Students need to be comfortable in finding the cube roots of values – a calculation that can be required in a number of topic areas and is regularly examined.

v.

$$P = 0.61 \times \pi \times r^2 \times w^3$$
$$= 0.61 \times \pi \times 43^2 \times 8^3$$
$$= 1\,814\,205.92 \text{ watts}$$

When speed of wind $\uparrow 10\%$

$$w' = 8 \times 110\% = 8.8 \text{ m/s}$$

Find P when $w' = 8.8$

$$P = 0.61 \times \pi \times 43^2 \times 8.8^3$$
$$= 2\,414\,708.08 \text{ watts}$$

Increase in Power = 2 414 708.08 – 1 814 205.92

= 600 502.16

$$\therefore \% \text{ Power increase} = \frac{600\,502.16}{1\,814\,205.92}$$
$$= 0.331$$
$$= 33\% \text{ (nearest \%)}$$

♦ Mean mark 41%
MARKER'S COMMENT: Students are reminded that a % increase requires them to find the difference in power generated at different wind speeds and divide this result by the original power output, as shown in the Worked Solution.

18. Algebra, STD2 A4 2004 HSC 26a

i. Percentage increase

$$= \frac{114 - 100}{100} \times 100$$

$$= 14\%$$

COMMENT: Common ADV/STD2 content in new syllabus.

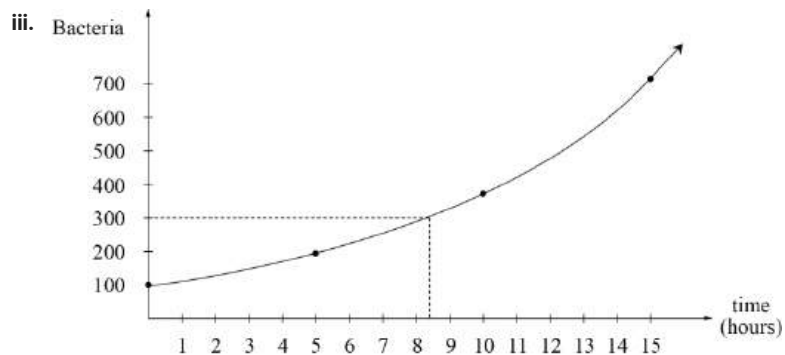
ii. $n = 100(1.14)^t$

When $t = 15$,

$$n = 100(1.14)^{15}$$

$$= 713.793 \dots$$

$$= 714 \text{ (nearest whole)}$$



iv. Using the graph

The number of bacteria reaches 300 after approximately 8.4 hours.

19. Algebra, STD2 A4 2021 HSC 24

a. Initial population occurs when $t = 0$.

$$P = 2000(1.2)^0$$

$$= 2000$$

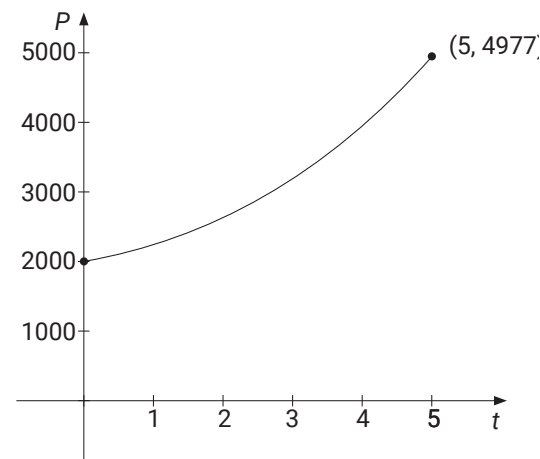
b. Find P when $t = 5$:

$$P = 2000(1.2)^5$$

$$= 4976.64$$

$$= 4977 \text{ (nearest whole)}$$

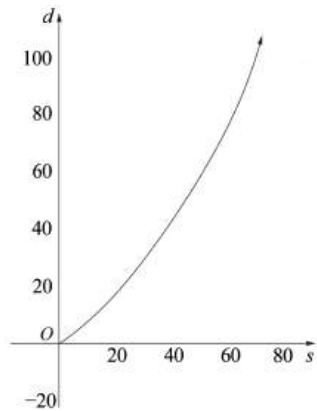
c.



♦ Mean mark (c) 48%.

20. Algebra, STD2 A4 2009 HSC 28a

(i)



(ii) When $s = 40$

$$\begin{aligned} d &= 0.01(40^2) + 0.7(40) \\ &= 16 + 28 \\ &= 44 \text{ m} \end{aligned}$$

When $s = 70$

$$\begin{aligned} d &= 0.01(70^2) + 0.7(70) \\ &= 49 + 49 \\ &= 98 \text{ m} \end{aligned}$$

$$\begin{aligned} \therefore \text{Difference} &= 98 - 44 \\ &= 54 \text{ metres} \end{aligned}$$

♦ Mean mark 41%
COMMENT: Students could easily have used the graph for calculating d at **40 km/h**, although the formula was required when the speed increased to **70 km/h**.

21. Algebra, STD2 A4 2017 HSC 28e

i. Graph is highest when increase = \$6

$$\begin{aligned} \therefore \text{Ticket price} &= 8 + 6 \\ &= \$14 \end{aligned}$$

♦ Mean mark 50%.

ii. Solution 1

$$\begin{aligned} \text{Tickets sold} &= 200 - (4 \times 10) \\ &= 140 \end{aligned}$$

Solution 2

$$\begin{aligned} \text{Tickets} &= \frac{\text{max income}}{\text{ticket price}} \\ &= \frac{1960}{14} \\ &= 140 \end{aligned}$$

♦ Mean mark 45%.

$$\begin{aligned} \text{iii. Cost} &= 140 \times \$2 + \$500 \\ &= \$780 \end{aligned}$$

$$\begin{aligned} \therefore \text{Profit when income is maximised} &= 1960 - 780 \\ &= \$1180 \end{aligned}$$

22. Algebra, STD2 A4 2009 HSC 28c

$$h \propto d^2$$

$$h = kd^2$$

$$\text{When } h = 1.6, d = 4.5$$

$$1.6 = k \times 4.5^2$$

$$\begin{aligned} \therefore k &= \frac{1.6}{4.5^2} \\ &= 0.07901 \dots \end{aligned}$$

Find h when $d = 15$

$$\begin{aligned} h &= 0.07901 \dots \times 15^2 \\ &= 17.777 \dots \\ &= 17.8 \text{ m (to 1 d.p.)} \end{aligned}$$

♦♦ Mean mark 22%
CRITICAL STEP: Reading the first line of the question carefully and establishing the relationship $h = kd^2$ is the key part of solving this question.

23. Algebra, STD2 A4 2019 HSC 31

a. The width of a rectangle cannot be negative.

◆ Mean mark 44%.

b. When $A = 18, w = 2$

Let $h =$ height of rectangle

$$18 = 2 \times h$$

$$h = 9 \text{ cm}$$

$$\begin{aligned}\therefore \text{Perimeter} &= 2 \times (2 + 9) \\ &= 22 \text{ cm}\end{aligned}$$

◆ Mean mark 42%.

24. Algebra, STD2 A4 2024 HSC 26

Graph cuts w -axis at $w = 0$ and $w = 40$.

By symmetry, maximum area occurs at $w = 20$.

$$A_{\max} = -0.5 \times 20^2 + 20 \times 20 = 200 \text{ cm}^2$$

$$A_{\max} = w \times h$$

$$200 = 20 \times h$$

$$h = 10 \text{ cm}$$

$$\therefore A_{\max} \text{ occurs when: } w = 20 \text{ cm, } h = 10 \text{ cm}$$

25. Algebra, STD2 A4 2020 HSC 19

a. From the graph, an area of $30\,000 \text{ m}^2$

can have an x -value of $x = 100$ or 200 m .

$$\therefore x_{\max} = 200 \text{ m}$$

◆ Mean mark part (a) 39%.

b. $A_{\max}$ occurs when $x = 150$

Substitute $x = 150$ into $3x + 2y = 900$:

$$3 \times 150 + 2y = 900$$

$$2y = 450$$

$$y = 225$$

$$\therefore \text{Maximum area when } x = 150 \text{ m and } y = 225 \text{ m}$$

◆◆ Mean mark part (b) 34%.

c. $A_{\max} = xy$

$$= 150 \times 225$$

$$= 33\,750 \text{ m}^2$$

◆ Mean mark part (c) 40%.

26. Algebra, STD2 A4 2020 HSC 33

a. When $n = 40$,

$$\text{Number of Bacteria } (y) = 4000$$

$$\text{b. } A = 1000 \Rightarrow y = 1000 b^n$$

By inspection, graph passes through $(40, 4000)$

$$\Rightarrow 4000 = 1000 b^{40}$$

Guess and check possible values of b :

$$\text{If } b = 1.03, \quad y = 1000 \times 1.03^{40} = 3262 \text{ (too low)}$$

$$\text{If } b = 1.04, \quad y = 1000 \times 1.04^{40} = 4801 \text{ (too high)}$$

$$\therefore 1.03 < b < 1.04$$

◆◆◆ Mean mark 14%.

27. Algebra, STD2 A4 2012 HSC 30b

i. Max height = 40m

COMMENT: With a mean mark of 92% in (i), a classic example of low hanging fruit in later questions.

ii. From graph

$$h = 35 \text{ when } x = 30 \text{ and } x = 170$$

$$\begin{aligned}\therefore \text{Horizontal distance} &= 170 - 30 \\ &= 140 \text{ m}\end{aligned}$$

iii. Ball hits ground at $x = 300$

$$\Rightarrow \text{Need to find } y \text{ when } x = 250$$

From graph, $y = 17.5 \text{ m}$ when $x = 250$

$$\therefore \text{Height of ball is } 17.5 \text{ m at a horizontal distance of } 50 \text{ m before } B.$$

MARKER'S COMMENT: Responses for (iii) in the range $17 \leq h \leq 18$ were deemed acceptable estimates read off the graph.

iv. Values of d not suitable.

If $d < 0$, it assumes the ball is hit away from point B . This is not the case in our example.

If $d > 300$, h becomes negative which is not possible given the ball cannot go below ground level.

◆◆◆ Mean mark (iv) 12%
MARKER'S COMMENT: Many students did not refer to the domain $d > 300$ as unsuitable to the model.

28. Algebra, STD2 A4 2012 HSC 30c

- i. 2030 occurs at $n = 20$ on the x -axis.

Expected population (no policy) = 6 600 000

- ii. A represents the population when $n = 0$ which is the population in 2010.

- iii. (1) $P = A(1.05)^n$ would be steeper and lie above
 $P = A(1.04)^n$ since $1.05 > 1.04$

- iii. (2) Let $b = 1.03$

$$\begin{aligned} P &= 3\,000\,000 \times 1.03^{20} \\ &= 5\,418\,000 \end{aligned}$$

Let $b = 1.02$

$$\begin{aligned} P &= 3\,000\,000 \times 1.02^{20} \\ &= 4\,457\,800 \end{aligned}$$

$$\therefore b = 1.02$$

- iv. In 2050, $n = 40$

$$\begin{aligned} P &= 3\,000\,000 \times 1.02^{40} \\ &= 6\,624\,119 \text{ (nearest whole)} \end{aligned}$$

Since the population is below 7 million,
the model will achieve the aim.

COMMENT: Common ADV/STD2 content in new syllabus.

♦ Mean mark part (ii) 48%

29. Algebra, STD2 A4 2006 HSC 28b

- i. 500 less vehicles per \$1 toll

$$12 \times 500 = 6000$$

$\therefore$ \$12 toll is the lowest for which no
vehicles will use the tunnel.

- ii. If the toll is \$5

$$5 \times 500 = 2500 \text{ less vehicles}$$

$$\begin{aligned} \therefore \text{Vehicles using the tunnel} \\ &= 6000 - 2500 \\ &= 3500 \end{aligned}$$

$$\begin{aligned} \therefore \text{Daily toll income} &= 3500 \times \$5 \\ &= \$17\,500 \end{aligned}$$

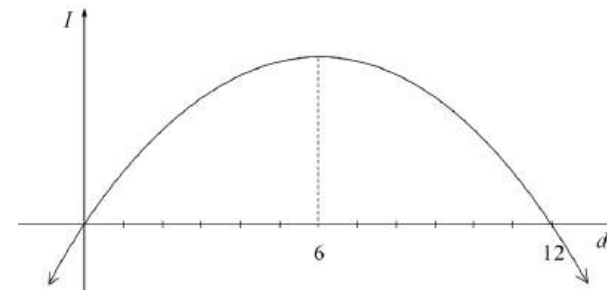
- iii. $d =$ toll

$v =$ Number of vehicles using the tunnel

$$\therefore v = 6000 - 500d$$

- iv. Income from tolls

$$\begin{aligned} &= \text{Number of vehicles} \times \text{toll} \\ &= (6000 - 500d) \times d \\ &= 6000d - 500d^2 \\ &= 500d(12 - d) \end{aligned}$$



From the graph, the maximum income from tolls
occurs when the toll is \$6.

$\therefore$ Anne is incorrect.

Alternate Solution

d	0	1	2	3	4	5
I	0	5500	10000	13500	16000	17500

d	6	7	8	12
I	18000	17500	16000	0

The table of values shows that income (I) increases and peaks when the toll hits \$6 before decreasing again as the toll gets more expensive.
 $\therefore$ Anne is incorrect.

30. Algebra, STD2 A4 2016 HSC 29b

- i. When $x = 0$,
- $$1.5 = A(1.1)^0$$
- $$\therefore A = 1.5 \text{ kg}$$
- ii. Daily growth rate
- $$= 0.1$$
- $$= 10\%$$

♦ Mean mark (i) 48%.
 ♦♦♦ Mean mark part (ii) 6%!

31. Algebra, STD2 A4 2015 HSC 29e

We can calculate the height of the board by finding the y -value at $t = 0$, which is 8 m.
 The diver’s maximum height occurs at $t = 0.5$, which is approximately 9.2 m.

$$\therefore \text{Maximum height above the board}$$

$$= 9.2 - 8$$

$$= 1.2 \text{ m}$$

♦♦ Mean mark 19%.

32. Algebra, STD2 A4 2021 HSC 35

- a. Highest revenue (R) occurs halfway between $x = -10$ and $x = 100$.

$$\text{Midpoint} = \frac{-10 + 100}{2} = 45$$

$$\therefore \text{Price of book for } R_{\text{max}}$$

$$= 45 + 10$$

$$= \$55$$

♦♦♦ Mean mark part (a) 16%.

- b. y -intercept $\rightarrow$ find R when $x = 0$:

$$R = -50(0)^2 + 4500(0) + 50\,000$$

$$= \$50\,000$$

♦♦♦ Mean mark part (b) 21%.